

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XVII

JEE (Main)-2022

TEST DATE: 14-07-2022

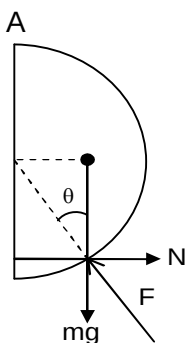
ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1.
Sol.

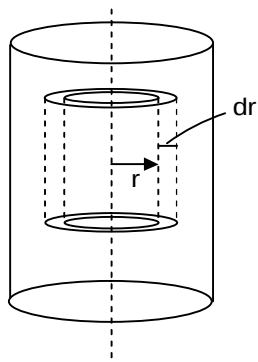


For equilibrium in this case all the forces should be passing through same point.

$$\therefore \tan \theta = \frac{1/2}{\sqrt{3}/2} = \frac{1}{\sqrt{3}}$$

2. D

Sol.



Using Gauss law for a thin cylinder gives

$$(E + dE)2\pi(r + dr)l - E2\pi rl = \frac{dq}{\epsilon_0}$$

$$\therefore \rho = \frac{dq}{2\pi r l dr}$$

$$= \epsilon_0 \left[\frac{dE}{dr} + \frac{E}{r} \right]$$

3. B

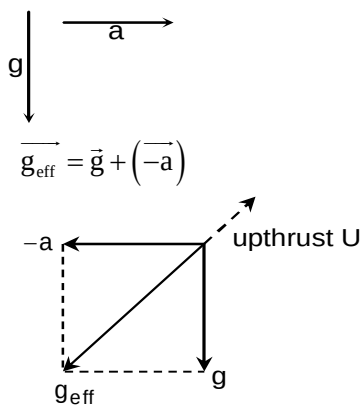
$$\text{Sol. } \frac{\theta_x}{\theta_y} = 2 \Rightarrow \frac{N_x A_x B_x}{N_y A_y B_y} = 2 \Rightarrow \frac{B_x}{B_y} = \frac{1}{2}$$

4. C

 Sol. Direction of magnetic field is along the direction of $\vec{V} \times \vec{E}$.

5. B

Sol.



Inclination will be in direction of U

6. C

$$\text{Sol. } P = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{10^{-14}}$$

$$KE = \frac{p^2}{2m}$$

$$KE = 8.2 \text{ MeV}$$

7. B

Sol. ${}^{32}\text{P} \rightarrow {}^{32}\text{S} + {}^0_1\text{e} + \vec{\nu}$
 $\Delta m = 2 \times 10^{-3} \text{ A.M.U}$
 $E = 1.862 \text{ MeV}$

8. A

Sol. $v = \omega A$ & $\omega = \frac{10}{2.5} = 4$

(A) $T = \frac{2\pi}{\omega} = \frac{\pi}{2} = 1.57$

(B) $a = \omega^2 A = 40$

(C) $v = \omega \sqrt{A^2 - x^2} = 2\sqrt{21}$

(D) $A = 2.5 \text{ cm}$

9. C

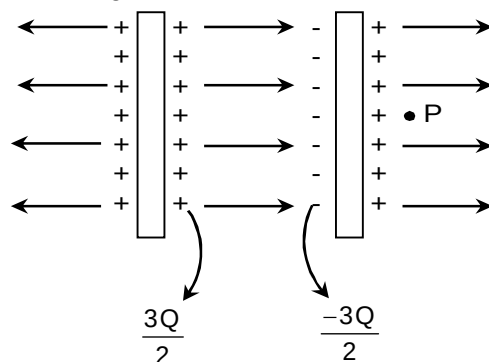
Sol. $\eta = \frac{w}{Q} = \frac{T_1 - T_2}{T_1}$

$\frac{w}{1000} = \frac{342}{184}$

$w = 500 \text{ J}$

10. D

Sol. Final charge on surfaces will be:



$\frac{Q_1 + Q_2}{2} = \frac{2Q - Q}{2} = \frac{Q}{2}$

Force between plates will be

$F = qE = \frac{Q^2}{2\pi\epsilon_0 d^2}$

11. D

Sol. $a = \frac{F - 5mg}{5m}$

$N - 3mg = 3\left(\frac{F - 5mg}{5m}\right)$

$\Rightarrow N = \frac{3F}{5}$

12. C

$$\text{Sol. } S_p = -\frac{(m_1 \overrightarrow{S_{1P}} + m_2 \overrightarrow{S_{2P}} + m_3 \overrightarrow{S_{3P}} + m_4 \overrightarrow{S_{4P}})}{m_1 + m_2 + m_3 + m_4 + m_p} = 0.25 \text{ m.}$$

13. B

$$\text{Sol. } L_{i\max} = \phi \quad \dots\dots(i)$$

$$q_{\max} = \omega i_{\max} \quad \dots\dots(ii)$$

$$\therefore q_{\max} = \phi \sqrt{\frac{C}{L}}$$

14. A

$$\text{Sol. } U_P = \frac{4}{3} \pi \left(\frac{D}{2}\right)^3 \rho g \times \frac{D}{2} = \frac{\pi D^4 g \rho}{12}$$

$$U_Q = D^3 \rho g \times \frac{D}{2} = \frac{D^4 \rho g}{2}$$

$$U_S = \pi \left(\frac{D}{2}\right)^2 D \rho g \frac{D}{2} = \frac{\pi D^4 \rho g}{8}$$

15. D

$$\text{Sol. } x = \frac{C}{T}$$

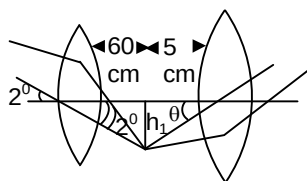
$$\frac{x_1}{x_2} = \frac{T_2}{T_1} = \frac{(273-173)}{(273-73)}$$

$$\frac{0.0040}{x_2} = \frac{100}{200}$$

$$x_2 = 0.0080$$

16. B

Sol.



$$2^\circ = \frac{h_1}{30} \quad \dots (1)$$

$$\theta = \frac{h_1}{5} \quad \dots (2)$$

$$\Rightarrow \frac{\theta}{2} = \frac{30}{5}$$

$$\theta = 12^\circ$$

17. C

Sol. Percentage error in measurement of unknown resistance is

$$\frac{dx}{x} = \frac{100}{\ell(100 - \ell)}$$

which is minimum when $\ell = 50$ cm.

18. B

Sol. $\text{Sinc} = \frac{\mu_1}{\mu_2}$

$$= \sqrt{\frac{K_1 M_{r1}}{K_2 M_{r2}}} = \frac{1}{\sqrt{2}}$$

$$C = 45^\circ$$

19. C

Sol. $P \propto T^2$

$$\Rightarrow V \propto \frac{1}{T}$$

$$nC_V(T_1 - T_2) = 300$$

$$n \times \frac{3}{2} R \left(T_1 - \frac{T_1}{3} \right) = 300$$

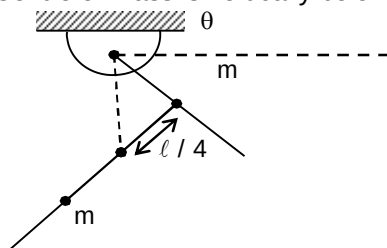
$$nRT_1 = 300$$

$$PV_0 = 300$$

$$P = 1.5 \times 10^5 \text{ Pa}$$

20. B

Sol. Centre of mass is vertically below the hinge.



$$\tan(90^\circ - \theta) = \frac{\ell/4}{\ell/2} = \frac{1}{2} \Rightarrow \tan \theta = 2$$

SECTION - B

21. 00026.25

Sol. $W_{\text{ext}} = W_s + Q_g$

$$= \frac{1}{2} K (\ell \sin \theta)^2 + mg \ell \sin \theta$$

$$= 26.25 \text{ J}$$

22. 00020.00

Sol. $SL_1 - SL_2 = 10 \log_{10} \left(\frac{I_{\text{max}}}{I_{\text{min}}} \right)$

$$\begin{aligned}
 &= 10 \log_{10} \left(\frac{A_1 + A_2}{A_1 - A_2} \right)^2 \\
 \Rightarrow SL_1 - SL_2 &= 20 \log_{10} \left[\frac{\frac{A_1}{A_2} + 1}{\frac{A_1}{A_2} - 1} \right] \\
 &= 20 \log_{10} 10 = 20 \text{ dB}
 \end{aligned}$$

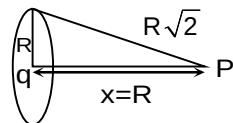
23. 00002.00

Sol. Electric field at point P due to charge of ring is:

$$E = \frac{kQx}{(R^2 + x^2)^{3/2}}$$

$$\text{At } x = R, E = \frac{kQ}{2\sqrt{2}R^2}$$

directed towards the center

 Electric field at P due to center charge is kq/R^2 for net field to be zero at P


$$\frac{kq}{R^2} = \frac{kQ}{2\sqrt{2}R^2} \quad \text{or} \quad q = \frac{Q}{2\sqrt{2}} = \frac{Q\sqrt{2}}{4}$$

24. 00002.50

$$\text{Sol. } \sigma = \frac{-\left(\frac{\Delta r}{r}\right)}{\left(\frac{\Delta \ell}{\ell}\right)}$$

$$\frac{\Delta A}{A} = \frac{2\Delta r}{r} = -2\% \quad \Rightarrow \left(\frac{\Delta r}{r}\right) = -1\%$$

$$\sigma = -\frac{\left(\frac{\Delta r}{r}\right)}{\left(\frac{\Delta \ell}{\ell}\right)}$$

$$(0.4) = -\frac{(-1\%)}{\left(\frac{\Delta \ell}{\ell}\right)} \quad \Rightarrow \left(\frac{\Delta \ell}{\ell}\right) = \frac{10}{4}\%$$

$$\left(\frac{\Delta \ell}{\ell}\right) = 2.5\%$$

25. 00992.50

$$\text{Sol. } i_g = 100 \times 10^{-4} = 0.01 \text{ A}$$

$$10 = 0.01 (R + 7.5) \Rightarrow R = 992.5 \Omega$$

26. 00000.80

Sol. $e = \frac{1.4 - 0.6}{1 - 0} = 0.8$

27. 00040.00

Sol. $\omega_r = \sqrt{\omega_1 \omega_2} = 40 \text{ rad/s}$

28. 00002.04

Sol. $t = (5.6 \text{ cm} - 8 \text{ VSD}) - (3.2 \text{ cm} - 4 \text{ VSD})$
 $= 2.4 \text{ cm} - 4 \times \frac{9}{10} \times 0.1$
 $= 2.04 \text{ cm}$

29. 00003.00

Sol. $V_1 = (1 + e)u_{cm} - eu_1$
 $= (1 + 0.75)10 - 0.75 \times 4 = 14.5 \text{ m/s}$

30. 00004.47

Sol. Taking point of projection as origin and x & y as horizontal and vertical,
 Equation of trajectory

$$y = x - \frac{1}{2}g \frac{x^2}{u^2} \times 2$$

$$y = x - g \frac{x^2}{u^2}$$

As it passes through $(-h, R)$

$$-h = R - g \frac{R^2}{u^2}$$

$$\Rightarrow u = \sqrt{\frac{gR^2}{h + R}} = 4.47 \text{ m/s}$$

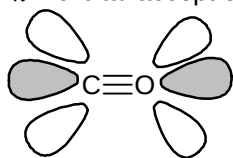
Chemistry

PART – B

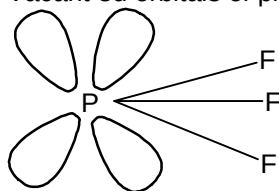
SECTION – A

31. B
Sol. (B) is only correct naming
(A) Incorrect because palatinate
(C) Ammine is missing

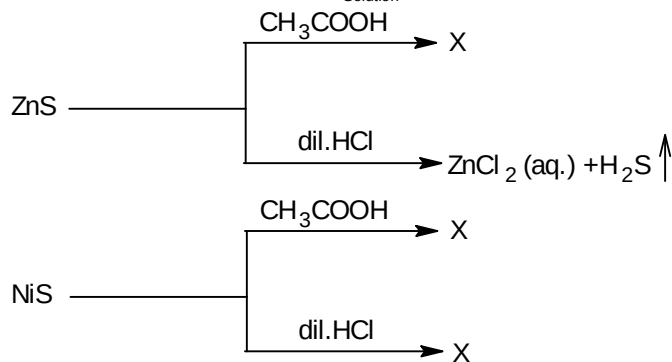
32. D
Sol. Both CO, PF₃ are non-classical (π -acceptor ligand)
 π^* -orbital accept electron from metal



Vacant 3d orbitals of phosphorus are accept electron from metal



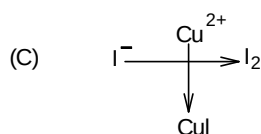
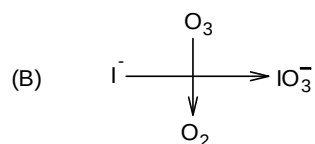
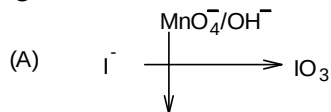
33. A
Sol. $\text{MnS} + 2\text{CH}_3\text{COOH} \rightarrow \text{Mn}(\text{CH}_3\text{COO})_2 + \text{H}_2\text{S} \uparrow$
(low K_{sp}) Solution



34. D
Sol. $\text{NaNO}_3 \xrightarrow[500^\circ\text{C}]{\Delta} \text{NaNO}_2 + \frac{1}{2}\text{O}_2$
- $\text{NO}_2^- + \text{NH}_2-\text{C}(=\text{S})-\text{NH}_2 \xrightarrow{\text{H}^+} \text{SCN}^- + \text{N}_2 + 2\text{H}_2\text{O}$
 (aq.)
- $\text{Fe}^{3+} + 3\text{SCN}^- \rightarrow \text{Fe}(\text{SCN})_3$ Blood red

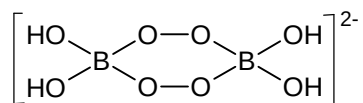
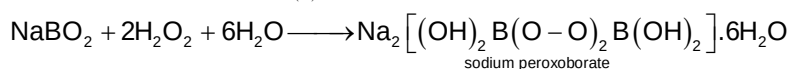
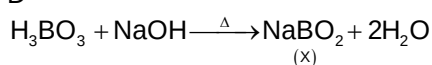
35. C

Sol.



36. D

Sol.



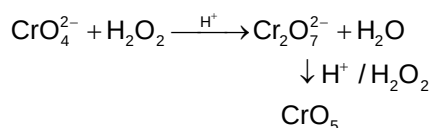
37. A

Sol.

In 6th period s-block element and Sc group element have no effect of lanthanide contraction because effect of lanthanide contraction is observed after Lu.

38. A

Sol.



CrO_5 is decompose to Cr^{3+} and O_2

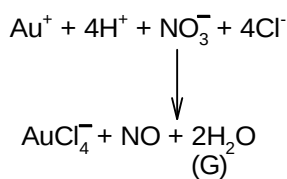
39. B

Sol.

M = Gold

Other metal = Cu

(A) + (B) = Aqua regia



40. B

Sol.

Theoretical

41. B

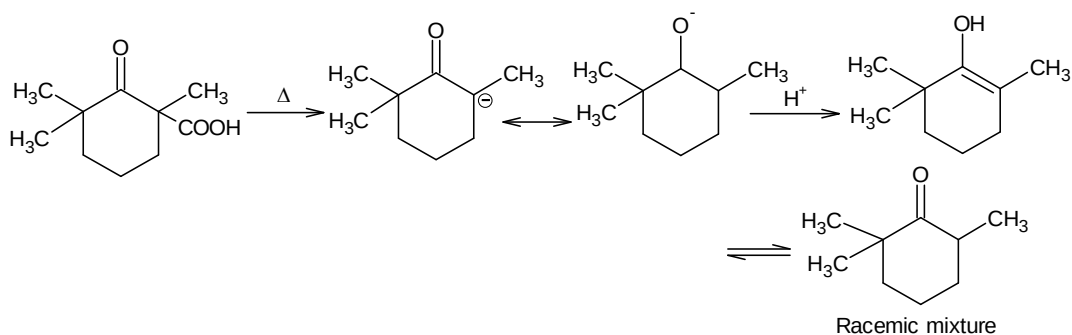
Sol.

Theoretical

42.

B

Sol. If strong electron withdrawing group is present at β position of carboxylic acid, then molecule losses CO_2 via formation of carbanion



43.

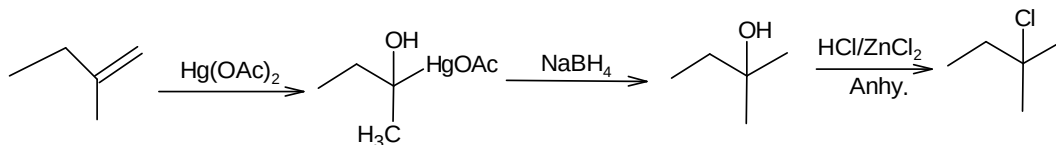
B

Sol. Halogenation in presence of light follows free radical substitution reaction. Greater is the stability of free radical, major is the corresponding product.

44.

D

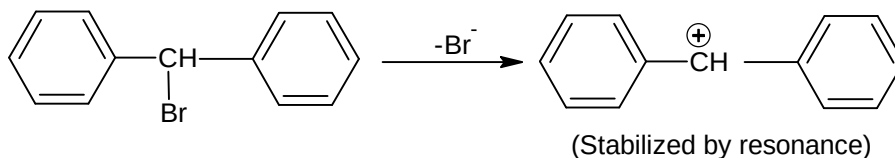
Sol.



45.

A

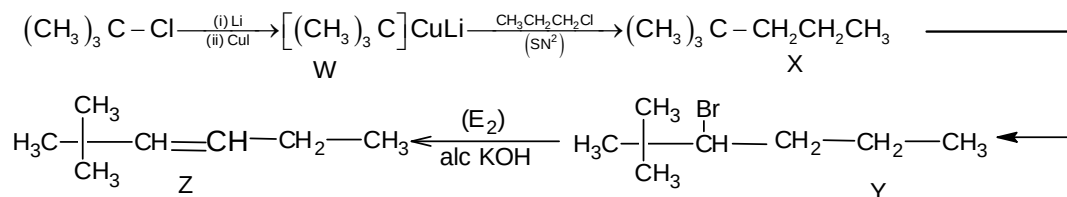
Sol. Rate of SN^1 reaction depends on stability of carbocation.



46.

D

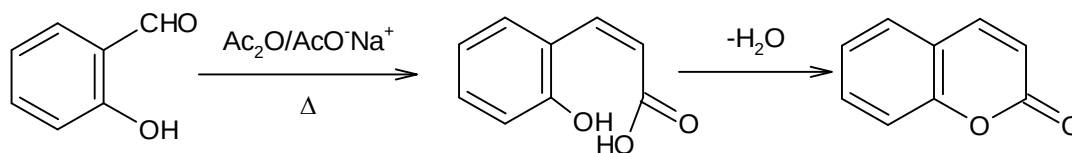
Sol.



47.

C

Sol.



48.

C

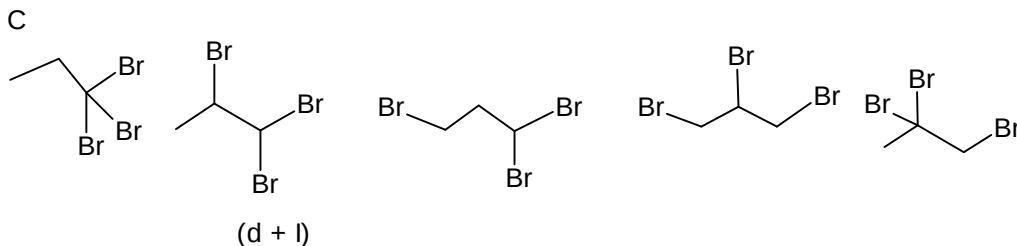
Sol. In Cannizzaro reaction best hydride donor is that which have attachment of more electron donating group.

49.

C

Sol. As molecule have $\text{DU} = 2$ so it cannot be an alkanolic acid.

50.
Sol.



SECTION – B

51. 02601.00

Sol. $\sqrt{v} = a(Z - b) = aZ - ab$
 Slope (a) = 1 ($\tan 45^\circ = 1$)
 Also given intercept (ab) = 1
 $\therefore \sqrt{v} = 1 \times 52 - 1 = 51$
 Or $v = 51^2 = 2601 \text{ s}^{-1}$

52. 00030.00

Sol. Electronic configuration of $_{17}\text{Cl} : 1s^2, 2s^2 2p^6 3s^2 3p^5$
 I excited state: $1s^2, 2s^2 2p^6, 3s^2 3p^4 3d^1$
 II excited state: $1s^2, 2s^2 2p^6, 3s^2 3p^3 3d^2$
 III excited state: $1s^2, 2s^2 2p^6, 3s^1 3p^3 3d^3$
 $\therefore (n+l) \text{ is : } [(3+0) + \{3 \times (3+1)\} + \{3 \times (3+2)\}]$
 $= 3 + 12 + 15 = 30$

53. 00089.40

Sol. $A(s) \rightleftharpoons B + D$
 $P_B + P_D = 40$
 $\therefore 2P = 40$
 Or $P = 20 \text{ atm}$
 $\therefore K_p = (P^2) = 20 \times 20 = 400$
 Similarly, $C(s) \rightleftharpoons E(g) + D(g)$
 $P_E + P_D = 80$
 $\therefore 2P = 80$
 Or $P = 40 \text{ atm}$
 $\therefore K_p = 40 \times 40 = 1600$
 For simultaneous equilibrium $A(s) \rightleftharpoons B + D$

$$\begin{array}{c} P_1 \quad P_1 + P_2 \\ C(d) \rightleftharpoons E + D \\ P_2 + P_2 + P_1 \end{array}$$

 $\therefore 400 = P_1 \times (P_1 + P_2)$
 $1600 = P_2 (P_1 + P_2)$
 Or $\frac{P_1}{P_2} = \frac{1}{4} \text{ or } P_2 = 4P_1$
 Also, $P_1(P_1 + P_2) = 400$

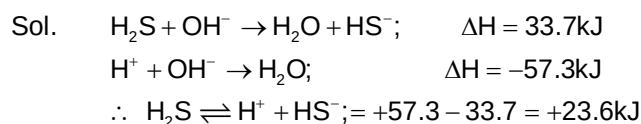
$$P_1(P_1 + 4P_1) = 400$$

$$5P_1^2 = 400$$

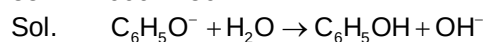
$$P_1 = 8.94 \text{ atm}$$

$$\begin{aligned} \text{Total pressure at eq.} &= P_1 + (P_1 + P_2) + P_2 \\ &= 2(P_1 + P_2) = 2(P_1 + 4P_1) \\ &= 10P_1 = 10 \times 8.94 = 89.4 \end{aligned}$$

54. 00023.60



55. 00011.50



$$[\text{OH}^-] = c \cdot h = c \cdot \sqrt{\frac{K_H}{c}} = \sqrt{\frac{K_w}{K_a}} \cdot c$$

$$\therefore \sqrt{K_b \cdot c} = \sqrt{10^{-4} \times 0.1} = 3.16 \times 10^{-3}$$

$$\therefore \text{pOH} = -2.5$$

$$\therefore \text{pH} = 11.5$$

56. 00263.12

Sol. For Liquid $\rightleftharpoons$ Vapour

$$\therefore \Delta H = T \Delta S$$

$$T = \frac{\Delta H}{\Delta S} = \frac{24.914 \times 10^3}{94.684} = 263.12 \text{ K}$$

57. 00029.10

Sol. $C_p = \frac{5}{2}R + R = \frac{7}{2}R = \frac{7 \times 8.314}{2}$
 $= 29.1 \text{ JK}^{-1} \text{ mol}^{-1}$

58. 00241.50

Sol. The number of units of AB in a unit cell (fcc) = 4
 Radius of cation $A^+ = 100 \text{ pm} = 100 \times 10^{-10} \text{ cm}$
 Radius of r^+ / r^- for octahedral coordination = 0.414
 Hence, Radius of the anion $(B^-) = \frac{100}{0.414} = 241.5 \text{ pm}$

59. 00048.30

Sol. Percentage of Cl

$$\begin{aligned} &= \frac{\text{Atomic mass of Cl} \times \text{mass of AgCl}}{\text{Molar mass of AgCl} \times \text{mass of}} \times 100 \\ &= \frac{35.5 \times 0.287 \times 100}{143.5 \times 0.147} = 48.3\% \end{aligned}$$

60. 00056.00

Sol. Meq. of H_2SO_4 taken $= 50 \times 0.5 \times 2 = 50$

Meq. of NaOH used after passing $\text{NH}_3 = 60 \times 0.5 = 30$

$\therefore$ Meq. of H_2SO_4 used for $\text{NH}_3 = 50 - 30 = 20 = (\text{NV})$

$$\begin{aligned}\therefore \quad \% \text{ of N} &= \frac{1.4 \times \text{NV}}{\text{mass of organic compound}} \\ &= \frac{1.4 \times 20}{0.5} = 56\%\end{aligned}$$

Mathematics

PART – C

SECTION – A

61. B

Sol. $\text{Sum} = {}^{51}C_1 + {}^{52}C_2 + {}^{53}C_3 + \dots + {}^{99}C_{49}$

$$= \text{coefficient of } x \text{ in } \left[(1+x)^{51} + x^{-1}(1+x)^{52} + x^{-2}(1+x)^{53} + \dots + x^{-48}(1+x)^{99} \right]$$

$$= \text{coefficient of } x \text{ in } \frac{(1+x)^{51} \left[\left(\frac{1+x}{x} \right)^{49} - 1 \right]}{\left(\frac{1+x}{x} - 1 \right)}$$

$$= \text{coefficient of } x \text{ in } \left[x^{-48}(1+x)^{100} - x(1+x)^{51} \right]$$

$$= {}^{100}C_{49} - 1 = {}^{100}C_{51} - 1$$

62. A

Sol. Clearly origin is the centroid, circumcentre and orthocentre, as given triangle is equilateral

$$\arg \left(\frac{z_2 - z_3}{2z_1 - z_2 - z_3} \right) = \arg \left(\frac{z_2 - z_3}{z_1 - \frac{z_2 + z_3}{2}} \right)$$

$$\arg \left(\frac{z_2 - z_3}{z_1 - \frac{z_2 + z_3}{2}} \right) = \pm \frac{\pi}{2}$$

$$\Rightarrow \frac{z_2 - z_3}{2z_1 - z_2 - z_3} \text{ is purely imaginary.}$$

63. D

Sol. $f(-x) \neq f(x)$ for the condition given in option (A)
 $f(-x) \neq -f(x)$ for the condition given in option (B)
 $f(x+T) \neq f(x)$ for the condition given in option (C)
Hence (D) is the correct option.

64. A

Sol. Pair of tangents from $(-2, -1)$ to the circle $x^2 + y^2 = 1$ is $(x^2 + y^2 - 1)(4) = (2x + y + 1)^2$
 $\Rightarrow 4xy - 3y^2 + 4x + 2y + 5 = 0$
 $\Rightarrow (y+1)(4x - 3y + 5) = 0$
Thus required equation is $4x - 3y + 5 = 0$

65. B

Sol. Let $\sin^{-1} 2x = t \Rightarrow x = \frac{1}{2} \sin t$

$$\Rightarrow dx = \frac{1}{2} \cos t dt$$

$$I = \int \frac{\frac{1}{2} \sin t + t^3}{\sqrt{1 - \sin^2 t}} \cdot \frac{1}{2} \cos t dt$$

$$= \frac{1}{4} \int \sin t dt + \frac{1}{2} \int t^3 dt = -\frac{1}{4} \cos t + \frac{1}{8} t^4 + k = -\frac{1}{4} \sqrt{1 - 4x^2} + \frac{1}{8} (\sin^{-1} 2x)^4 + k.$$

66. B

Sol. Equation of chord of contact with respect to point $(-4, 2)$ is $-\frac{4x}{a^2} - \frac{2y}{b^2} = 1$ and with respect to point $(2, 1)$ is $\frac{2x}{a^2} - \frac{y}{b^2} = 1$.

$$\text{According to given condition } \begin{pmatrix} \frac{4}{a^2} \\ -\frac{2}{b^2} \end{pmatrix} \times \begin{pmatrix} -\frac{2}{a^2} \\ \frac{1}{b^2} \end{pmatrix} = -1 \Rightarrow \frac{b^4}{a^4} = \frac{1}{4}.$$

$$\therefore e = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{1 + \frac{1}{2}} = \sqrt{\frac{3}{2}}.$$

67. B

Sol. $\frac{dy}{dx} = (1 - 2x)$

$$y = x - x^2 + c$$

As it passes through $(2, -2)$

$$\therefore c = 0$$

Curve is $y = x - x^2$

According to question

$$\frac{32}{3} = \int_0^{1-\alpha} [(x - x^2) - \alpha x] dx$$

$$\frac{32}{3} = \left[(1-\alpha) \frac{x^2}{2} - \frac{x^3}{3} \right]_0^{1-\alpha}$$

$$\Rightarrow \frac{32}{3} = \frac{(1-\alpha)^3}{2} - \frac{(1-\alpha)^3}{3}$$

$$\Rightarrow (1-\alpha)^3 = 64$$

$$\Rightarrow 1-\alpha = 4 \quad \therefore \alpha = -3.$$

68. B

Sol. $2\pi = 5 \tan^{-1}(x^2 + x + a) + 3 \cot^{-1}(x^2 + x + a) = \frac{3\pi}{2} + 2 \tan^{-1}(x^2 + x + a)$

$$\Rightarrow x^2 + x + a = 1. \text{ For required condition } 1 - 4(a - 1) = 0 \Rightarrow a = \frac{5}{4}.$$

69. B

 Sol. (A) is not true, for example take $f(x) = \cos x$, $g(x) = \sin x$, $a = 0$, $b = \frac{\pi}{4}$, (C) is not true,

 for let $f(x) = -1$, $g(x) = -2$. We know that if $f(x) > 0$, then $\int_a^b f(x) dx > 0$,

 thus $\int_a^b (f(x) - g(x)) dx > 0$.

Hence (B) is the correct choice.

70. B

 Sol. Let $\vec{c} = \lambda \vec{a} + \mu \vec{b}$

$$\text{Now } 0 = \vec{c} \cdot \vec{b} = \lambda \vec{a} \cdot \vec{b} + \mu b^2 = -\lambda + 5\mu \quad (1)$$

$$\text{Again } 7 = \vec{a} \cdot \vec{c} = \lambda a^2 + \mu \vec{a} \cdot \vec{b} = 3\lambda - \mu \quad (2)$$

$$\text{Solving (1) and (2), } \lambda = \frac{5}{2}, \mu = \frac{1}{2}$$

$$\text{Thus } \vec{c} = \frac{5}{2}(-\hat{i} + \hat{j} + \hat{k}) + \frac{1}{2}(2\hat{i} + \hat{k}) = -\frac{3}{2}\hat{i} + \frac{5}{2}\hat{j} + 3\hat{k}$$

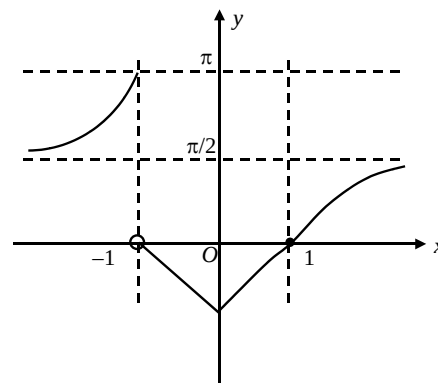
71. B

Sol. Whenever one player is knocked out exactly one match is played and vice versa.

 If number of players is n , number of players to be knocked out is $(n - 1)$.

 $\therefore$ Number of matches played = $n - 1$.

72. B

 Sol. From the figure, it is obvious that f is discontinuous at $x = -1$.


73. B

 Sol. f is continuous at $x = 0$

 Further $f(0 + h) > f(0)$ and $f(0 - h) > f(0)$, for positive h . Hence f has minimum value at $x = 0$.

74. B

$$\text{Sol. } \Delta = \begin{vmatrix} b+c-2a & c+a-2b & a+b-2c \\ bc & ca & ab \\ a^2 & b^2 & c^2 \end{vmatrix} = \begin{vmatrix} a+b+c-3a & a+b+c-3b & a+b+c-3c \\ bc & ca & ab \\ a^2 & b^2 & c^2 \end{vmatrix}$$

$$= \begin{vmatrix} a+b+c & a+b+c & a+b+c \\ bc & ca & ab \\ a^2 & b^2 & c^2 \end{vmatrix} - \begin{vmatrix} 3a & 3b & 3c \\ bc & ca & ab \\ a^2 & b^2 & c^2 \end{vmatrix}$$

$$= (a+b+c) \begin{vmatrix} 1 & 1 & 1 \\ bc & ca & ab \\ a^2 & b^2 & c^2 \end{vmatrix} - 3 \begin{vmatrix} a & b & c \\ bc & ca & ab \\ a^2 & b^2 & c^2 \end{vmatrix}$$

Multiplying the 3 columns by a, b, c respectively and taking abc common from 2nd row in both the determinants

$$= (a+b+c) \begin{vmatrix} a & b & c \\ 1 & 1 & 1 \\ a^3 & b^3 & c^3 \end{vmatrix} - 3 \begin{vmatrix} a^2 & b^2 & c^2 \\ 1 & 1 & 1 \\ a^3 & b^3 & c^3 \end{vmatrix}$$

$$= -(a+b+c) \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} + 3 \begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix}$$

$$= -(a+b+c)(a-b)(b-c)(c-a)(a+b+c) + 3(a-b)(b-c)(c-a)(ab+bc+ca)$$

$$= -(a-b)(b-c)(c-a)[(a+b+c)^2 - 3(ab+bc+ca)]$$

$$= -(a-b)(b-c)(c-a)[a^2+b^2+c^2-ab-bc-ca]$$

$$= -(a-b)(b-c)(c-a) \left(\frac{(a-b)^2 + (b-c)^2 + (c-a)^2}{2} \right) = 0$$

$\Rightarrow$ Either $a = b$ or $b = c$ or $c = a$ or $a = b = c$

75. D

Sol. $\vec{a} \cdot \vec{b} = 0$

$$\vec{c} \cdot \vec{a} = \vec{c} \cdot \vec{b} = \cos \theta$$

$$\vec{c} \cdot \vec{a} = \lambda(\vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b}) + \mu \vec{a} \cdot (\vec{a} \times \vec{b})$$

$$\cos \theta = \lambda$$

$$\vec{c} \cdot \vec{c} = \lambda^2(\vec{a} + \vec{b})^2 + (1.1 \cdot \sin 90^\circ)^2 + 2\lambda\mu\{(\vec{a} + \vec{b}) \cdot (\vec{a} \times \vec{b})\}$$

$$1 = \lambda^2(1 + 1 + 2.0) + \mu^2 + 2\lambda\mu(0) \quad [\because |\vec{a}| = 1, |\vec{b}| = 1 \text{ and } \vec{a} \cdot \vec{b} = 0]$$

$$1 = 2\cos^2\theta + \mu^2$$

$$\mu^2 = 1 - 2\cos^2\theta \geq 0$$

$$-\frac{1}{\sqrt{2}} \leq \cos \theta \leq \frac{1}{\sqrt{2}} \Rightarrow \theta \in \left[\frac{\pi}{4}, \frac{3\pi}{4} \right].$$

76. B

Sol. $\frac{\frac{b}{2} - a}{\sqrt{14}} = \frac{\frac{c}{3} - \frac{b}{2}}{\sqrt{14}}$

$$\begin{aligned}\Rightarrow b &= a + \frac{c}{3} \\ \Rightarrow 6b &= 6a + 2c \\ \Rightarrow 6a, 3b, 2c &\text{ are in A.P.}\end{aligned}$$

77. C

Sol. $(xy \cos x - y \sin x)dx = x \sin x dy$

$$\begin{aligned}\Rightarrow \frac{\cos x}{\sin x} dx &= \frac{y dx + x dy}{xy} \\ \Rightarrow \ln |\sin x| &= \ln |xy| + \log C \\ \Rightarrow \frac{\sin x}{xy} &= C\end{aligned}$$

78. B

Sol. Slope of tangent at $P(x_1, y_1) \equiv \left(\frac{dy}{dx}\right)_P = 2x_1$

$$\Rightarrow \text{slope of normal at } P \text{ is } = -\frac{1}{2x_1} = -\frac{a}{b} \Rightarrow x_1 = \frac{b}{2a}$$

$$\Rightarrow y_1 = \frac{b^2}{4a^2} \text{ and } ax_1 + by_1 + c = 0$$

$$\Rightarrow b^3 + 2a^2b + 4a^2c = 0.$$

Hence $x = b$ is a real root of $x^3 + 2a^2x + 4a^2c = 0$

Let $f(x) = x^3 + 2a^2x + 4a^2c \Rightarrow f'(x) = 3x^2 + 2a^2 > 0$

If $a \neq 0 \Rightarrow f(x) = 0$ has exactly one real root i.e., $x = b$.

79. C

$$\begin{aligned}\text{Sol. } (\bar{x}) &= \frac{\text{sum of quantities}}{n} = \frac{\frac{n}{2}(a + \ell)}{n} \\ &= \frac{1}{2}[1 + 1 + 100d] = 1 + 50d\end{aligned}$$

$$\therefore MD = \frac{1}{n} \sum |x_i - \bar{x}| \Rightarrow 255$$

$$= \frac{2d}{101} \left(\frac{50 \times 51}{2} \right)$$

$$\therefore d = \frac{255 \times 101}{51 \times 50} = 10.1$$

80. A

Sol. Suman is brilliant and dishonest if and only if Suman is rich, is expressed as,

$$Q \leftrightarrow (P \wedge \sim R)$$

So, negation of it will be, $\sim [Q \leftrightarrow (P \wedge \sim R)]$.

SECTION – B

81. 00032.00

Sol. $|A| = 5, |4A| = 4^3 \cdot |A| = 64(5) = 320$

82. 00001.00

Sol. $\log_{10}(2^{x+1} + x - 37) = x \log_{10} 2$

$$\Rightarrow x = 37 - 2^x$$

$$\Rightarrow x = 5$$

83. 00014.00

Sol. Assume $x^2 - 6x + 6 = x^2 - 4x + 4$

$$\Rightarrow x = 1$$

substituting $x = 1$ in given functional relation

$$\Rightarrow 2f(1) = 2 \Rightarrow f(1) = 1$$

$$\text{for } x^2 - 6x + 6 = 1$$

$$\Rightarrow x = 1 \text{ to } 5$$

$$x^2 - 4x + 4 = 1 \Rightarrow x = 1 \text{ or } 3$$

$$\text{putting } x = 3 \text{ and } 5 \text{ in } f(x^2 - 6x + 6) + f(x^2 - 4x + 4) = 2x$$

$$\text{we get } f(-3) + f(1) = 6$$

$$\Rightarrow f(-3) = 5 \text{ and } f(1) + f(9) = 10 \Rightarrow f(9) = 9$$

$$\Rightarrow f(-3) + f(9) = 14$$

84. 00001.00

Sol. $a^2 + b^2 \geq 1$

85. 00168.00

Sol. $2A^{-1} = \frac{\text{adj}(2A)}{|2A|} = 2^{3-1} \frac{\text{adj}A}{2^3 |A|} = \frac{\text{adj}A}{14}$

$$\Rightarrow |2A^{-1}| = \frac{1}{14^3} |\text{adj}A| = \frac{1}{14^3} |A|^{3-1} = \frac{1}{14^3} \times 7^2 = \frac{1}{56}$$

86. 00021.00

Sol. $B \quad B \quad B \quad B \quad B \quad B \quad B$

$$|W| \quad |W| \quad |W| \quad |W| \quad |W| \quad |W| \quad |W|$$

$$n(s) = \frac{10!}{7! 3!}$$

$$n(E) = {}^8C_3 = \frac{8!}{3! 5!}, \text{ because there are 8 places for 3 black balls}$$

$$\therefore P(E) = \frac{\frac{8!}{3! 5!}}{\frac{10!}{7! 3!}} = \frac{8! 7!}{10! 5!} = \frac{7.6}{10.9} = \frac{7}{15}$$

87. 00303.00

Sol. 7^α always ends with 7, 9, 3, 1

$$\therefore 7^\alpha + 7^\beta \text{ is divisible by 5}$$

$$\Rightarrow \text{favourable events} = (25 \times 25) \times 2$$

sample space = 100×100

$$\therefore \text{required prob.} = \frac{2 \times (25 \times 25)}{100 \times 100} = \frac{1}{8}$$

88. 00000.00

Sol. $\therefore$ 1 lies between roots of $y^2 - my + 1 = 0$

$$\Rightarrow f(1) < 0 \Rightarrow m > 2$$

$$\text{Let } z = \frac{4|x|}{x^2 + 16} = \frac{4t}{t^2 + 16} \quad \forall t = |x|$$

$$\therefore zt^2 - 4t - 16z = 0$$

$$\therefore t \geq 0$$

$$0 \leq z \leq \frac{1}{2}$$

$$\Rightarrow [z^m] = 0 \quad \forall m > 2.$$

89. 00013.00

Sol. Expression = $(x \cdot x^2 \cdot x^3 \dots x^{20}) \cdot E = x^{210} E$
where

$$E = \left(1 - \frac{1}{x}\right) \left(1 - \frac{2}{x^2}\right) \left(1 - \frac{3}{x^3}\right) \dots \left(1 - \frac{20}{x^{20}}\right)$$

$$\text{Coefficient of } x^{-7} \text{ in } E \\ = -7 + 6 + 10 + 12 - 8 = 13.$$

90. 00000.00

Sol. $\cos x - \sin x = \operatorname{cosec} x$

$$\Rightarrow \cos x = \sin x + \frac{1}{\sin x}$$

$$\text{Now } \left| \sin x + \frac{1}{\sin x} \right| \geq 2 \text{ but } |\cos x| \leq 1$$

So no solution.